

CSC 514 – COMPUTER GRAPHICS

Rain Semester Examination 2023/2024 (July 2025) Full Answer Guide

Grounded in the uploaded CSC 514 notes, glossary and model-solution files.

SECTION A – Fill in the Gaps (20 marks)

1. Smallest size object that can be displayed on a monitor is called **PIXEL** (picture element).
2. The translation distances (dx, dy) are called **TRANSLATION VECTOR** (shift vector).
3. The CMY model is converted to CMYK by adding a _____ component **BLACK / KEY (K)** component.
4. CMY is based on _____ mixing, where combining all primary colours ideally results in _____ **SUBTRACTIVE** mixing; **BLACK**.
5. `glEnable(GL_CULL_FACE)` helps performance by skipping **BACK-FACING POLYGONS** (back faces).
6. `glViewport(0,0,width,height)` defines the portion of the _____ where rendering will appear **WINDOW** (screen/framebuffer area).
7. Shearing is represented by non-zero entries in the _____ positions of the transformation matrix **OFF-DIAGONAL** positions.
8. CMY is based on _____ mixing, where combining all primary colours ideally results in _____ **SUBTRACTIVE** mixing; **BLACK**.
9. A point (4.2, 3.8) can be displayed on a screen by pixel (4, 4).
10. Texture mapping occurs in the graphics pipeline at the **FRAGMENT PROCESSING STAGE** (after rasterisation / fragment shader).
11. Computer graphics allows _____ on pictures before displaying them **CREATION, STORAGE, MANIPULATION, TRANSFORMATION** and **RETRIEVAL**.
12. Primary use of clipping in computer graphics **DISCARD** the parts of objects outside the clip window/viewing region so only visible geometry is rendered.
13. A bitmap is a collection of _____ that describes an image **BITS** (pixel values).
14. OpenGL primitives are constructed from _____ and examples include **GL_POINTS, GL_LINES** and _____ **VERTICES; GL_TRIANGLES**.
15. `viewport(0,0,width,height)` defines the portion of the _____ where rendering will appear **WINDOW**.
16. Shearing transformations are represented by non-zero entries in the _____ positions **OFF-DIAGONAL** positions.
17. The basic geometric structures that describe a scene on display are called **OUTPUT PRIMITIVES**.
18. _____ is responsible for accessing the frame buffer to refresh the screen **VIDEO CONTROLLER** (display controller).
19. Vector graphics is composed of **PATHS** (mathematically defined lines and curves).
20. The point about which an object is rotated is called **PIVOT POINT / CENTRE OF ROTATION**.

SECTION B

Question 1 (15 marks)

1(a) What is Computer Graphics?

Computer graphics is the field concerned with the creation, storage, manipulation, transformation, retrieval and display of pictures or visual information using a computer.

1(b) Applications of Computer Graphics

- **Entertainment:** Used in films, animation, television, games and visual effects to create and display realistic or imaginary scenes.
- **Military:** Used in flight/weapon simulators, battlefield visualisation, map displays, mission planning and training environments.
- **Simulation:** Used to imitate real systems visually, such as aircraft, driving, industrial processes and scientific experiments, without using the real system.
- **Forensics:** Used to reconstruct crime/accident scenes, enhance visual evidence, create facial reconstructions and present spatial evidence.
- **Medicine:** Used to display and reconstruct CT, MRI, ultrasound and X-ray data for diagnosis, treatment planning and surgical simulation.

1(c) Raster Graphics vs Vector Graphics

Basis	Raster Graphics	Vector Graphics
Basic element	Pixels	Paths: points, lines, curves and polygons
Resolution	Resolution-dependent	Resolution-independent
Scaling	May become pixelated/jagged	Scales without loss of quality
Best for	Photographs, video, realistic images	Logos, CAD, diagrams, illustrations
Storage	Stores pixel values	Stores mathematical drawing instructions

1(d) Hardware Input Devices Used for Computer Graphics

- Keyboard – enters commands, numbers and text.
- Mouse – selects, drags, draws and manipulates objects using a pointer.
- Graphics tablet/digitizer – allows accurate free-hand drawing using a stylus.
- Joystick – provides directional control, useful for games and interactive 3D navigation.
- Light pen/stylus – points to or draws directly on a compatible display/tablet.
- Scanner/digital camera – digitizes existing photographs and drawings into computer-readable images.
- Touch screen – accepts direct finger or stylus input on the display.

Question 2 (15 marks)

2(a) Additive and Subtractive Colour Mixing

Feature	Additive Mixing	Subtractive Mixing
Primaries	Red, Green, Blue (RGB)	Cyan, Magenta, Yellow (CMY/CMYK)
How colour is made	Light is added together	Ink/pigment absorbs (subtracts) light
Starting point	Black (absence of light)	White paper/light
All primaries combined	White	Ideally black
Devices	Monitors, TVs, projectors	Printers, plotters, hardcopy

2(b) Terms in Computer Graphics

Pixel: The smallest addressable picture element on a raster display.

Rendering: The process of generating the final 2D image from a scene description, geometry, viewpoint, lighting, textures and materials.

Coordinate System: A reference framework used to specify the numerical position of points, usually with x and y in 2D and x, y, z in 3D.

Digitizing: Converting a continuous/analogue picture or object into a discrete digital representation that a computer can store and process.

Dithering: Creating the illusion of unavailable colours or intensity levels by arranging available pixel colours in patterns that the eye blends.

2D Graphics: Graphics represented on a two-dimensional plane using x and y coordinates.

3D Graphics: Graphics represented in three-dimensional space using x, y and z coordinates, allowing depth and perspective.

2(c) Output and Geometric Primitives

Output primitives: the basic geometric structures used to describe a scene on a display, such as points, lines and polygons.

Geometric primitives: fundamental building blocks of geometry such as points, lines/line segments, triangles, circles, spheres and polygons.

- **Line:** A straight geometric primitive joining two endpoints; in raster graphics it is scan-converted into a sequence of pixels.
- **Circle:** The locus of points at constant distance r from a centre (x_c, y_c) : $(x-x_c)^2 + (y-y_c)^2 = r^2$.
- **Polygon:** A closed figure formed by a sequence of line segments (edges) joined end-to-end at vertices.

2(d) Homogeneous 2D and 3D Transformations

Homogeneous coordinates add one extra coordinate so that translation, rotation, scaling and other affine transformations can all be represented by matrix multiplication. A 2D point (x,y) becomes $(x,y,1)$; a 3D point (x,y,z) becomes $(x,y,z,1)$.

2D translation example:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Example: translating by $(5,1)$ gives $x' = x+5$ and $y' = y+1$.

3D homogeneous translation:

$$\begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The extra homogeneous coordinate lets graphics hardware combine several transformations into one composite matrix.

Question 3 (15 marks)

3(a) OpenGL program fragment to open a window and draw a square

```
#include <GL/glut.h>

void display(void)
{
    glClear(GL_COLOR_BUFFER_BIT);
    glColor3f(1.0, 0.0, 0.0);

    glBegin(GL_QUADS);
        glVertex2f(-0.5, -0.5);
        glVertex2f( 0.5, -0.5);
        glVertex2f( 0.5,  0.5);
        glVertex2f(-0.5,  0.5);
    glEnd();

    glFlush();
}

int main(int argc, char **argv)
{
    glutInit(&argc, argv);
    glutCreateWindow("Square");
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}
```

This fragment uses the OpenGL/GLUT ideas in the course material: create a window, begin a primitive, specify its vertices, end the primitive and display it.

3(b) OpenGL commands

- Straight lines: `glBegin(GL_LINES); ... glVertex... ; glEnd();`
- Triangles: `glBegin(GL_TRIANGLES); ... three vertices per triangle ... glEnd();`
- A point: `glBegin(GL_POINTS); glVertex... ; glEnd();`

RGB colour values using `glColor3f(R,G,B)`:

Colour	<code>glColor3f</code> value
Red	(1,0,0)
Green	(0,1,0)
Blue	(0,0,1)
Yellow	(1,1,0)
Magenta	(1,0,1)
Cyan	(0,1,1)
White	(1,1,1)
Black	(0,0,0)

3(c) OpenGL API and five primitives

OpenGL is a graphics Application Programming Interface (API): a standard set of commands/functions used by programs to communicate with graphics hardware and render images.

- `GL_POINTS`
- `GL_LINES`
- `GL_LINE_STRIP`
- `GL_LINE_LOOP`
- `GL_TRIANGLES`

3(d) Difference between GL_LINES and GL_LINE_STRIP

GL_LINES: Treats each pair of vertices as an independent line segment. Vertices 1–2 form one line, 3–4 another, etc.

GL_LINE_STRIP: Draws a connected open polyline. Each new vertex is connected to the previous one; n vertices produce n–1 connected segments.

Question 4 (15 marks) – Transformations

4(a) 2D rotation matrix for 90° clockwise

General anticlockwise rotation matrix:

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

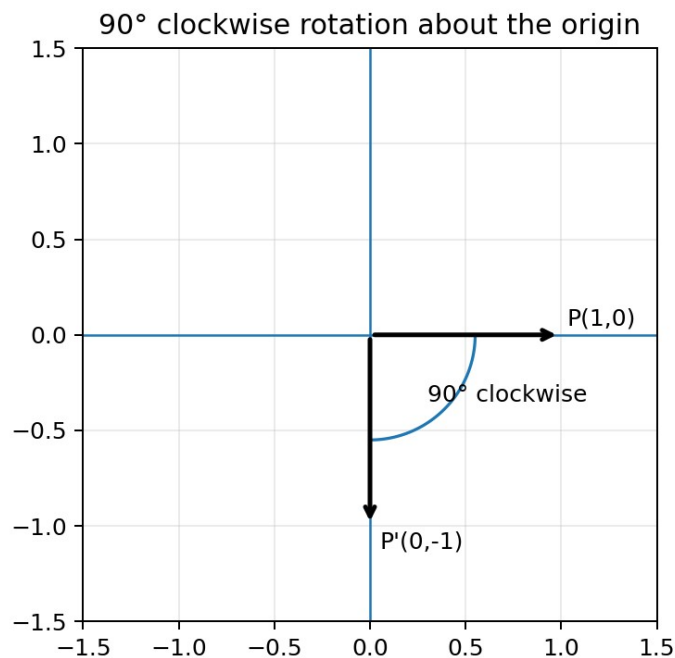
Clockwise 90° means $\theta = -90^\circ$. Therefore $\cos(-90^\circ)=0$ and $\sin(-90^\circ)=-1$.

$$R(-90^\circ) = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

Homogeneous 3×3 form:

$$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Check: $(1,0) \rightarrow (0,-1)$, which is a quarter-turn clockwise.



4(b)(i) Rotation of an object by 60° about the origin

No direction is stated, so the standard positive (anticlockwise) convention is used.

$\cos 60^\circ = 0.5$ and $\sin 60^\circ = 0.8660$

$$R(60^\circ) = \begin{bmatrix} 0.5000 & -0.8660 \\ 0.8660 & 0.5000 \end{bmatrix}$$

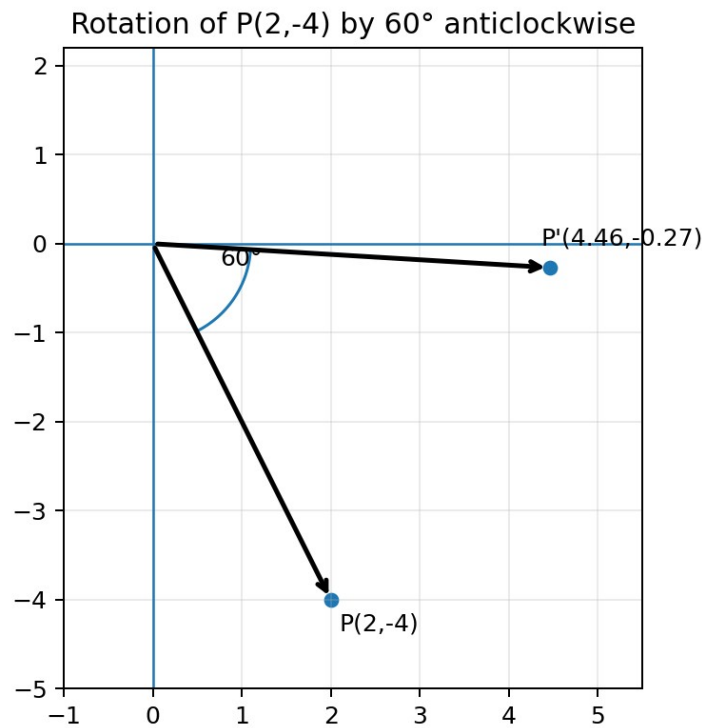
4(b)(ii) New coordinates of P(2,-4)

Use $x' = x \cos \theta - y \sin \theta$ and $y' = x \sin \theta + y \cos \theta$.

$$\begin{aligned}x' &= (2)(0.5) - (-4)(0.8660) \\&= 1 + 3.4641 \\&= 4.4641\end{aligned}$$

$$\begin{aligned}y' &= (2)(0.8660) + (-4)(0.5) \\&= 1.7321 - 2 \\&= -0.2679\end{aligned}$$

FINAL ANSWER: $P' = (4.4641, -0.2679) \approx (4.46, -0.27)$



4(c) Scaling with respect to the origin

(i) n units in X only:

$$\begin{bmatrix} n & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = nx, y' = y$$

(ii) m units in Y only:

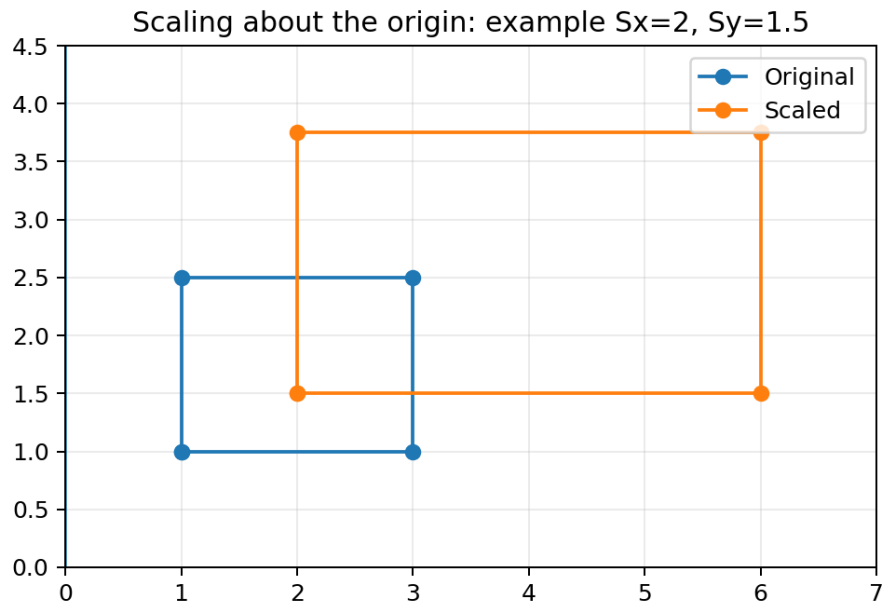
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = x, y' = my$$

(iii) n in X and m in Y simultaneously:

$$\begin{bmatrix} n & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x' = nx, y' = my$$



4(d) 3×3 matrix translating (x,y) 5 units right and 1 unit up

For translation, $t_x=+5$ and $t_y=+1$:

$$M_1 = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Verification:

$$\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x+5 \\ y+1 \\ 1 \end{bmatrix}$$

FINAL: the point moves 5 units to the right and 1 unit upward.

Question 5 (15 marks) – Colour Models

5(a) Digital printer uses CMY; computer screen uses RGB

RGB is additive and used by light-emitting displays. CMY is subtractive and used by printers/hardcopy devices. The models are complementary on the normalised [0,1] scale:

$$\begin{aligned} C &= 1 - R \\ M &= 1 - G \\ Y &= 1 - B \end{aligned}$$

5(a)(i) Convert RGB (0.8, 0.1, 0.3) to CMY

$$\begin{aligned} C &= 1 - 0.8 = 0.2 \\ M &= 1 - 0.1 = 0.9 \\ Y &= 1 - 0.3 = 0.7 \end{aligned}$$

FINAL ANSWER: CMY = (0.2, 0.9, 0.7)

5(b) Pseudocode for RGB → CMY

```
ALGORITHM RGB_to_CMY
INPUT: R, G, B
OUTPUT: C, M, Y
BEGIN
```

```

IF R > 1 OR G > 1 OR B > 1 THEN
    R = R / 255
    G = G / 255
    B = B / 255
END IF

C = 1 - R
M = 1 - G
Y = 1 - B

RETURN (C, M, Y)
END

```

5(c) Why printed colour may differ from the screen

- RGB forms colour by emitted light, whereas CMY forms colour by absorbing/reflected light.
- The screen and printer have different colour gamuts; some vivid screen colours cannot be reproduced by ink.
- RGB and CMY are device-dependent, so different monitors and printers can reproduce the same numeric values differently.
- Real printing inks are not perfect mathematical complements.
- CMY does not produce a perfect black in practice, so printers add a black K channel (CMYK).
- Paper, ink absorption/dot gain, ambient lighting, calibration, gamma and resolution affect the appearance.

5(d) Direct RGB coding with 10 bits per primary

Each primary has 10 bits:

$$\text{Levels per primary} = 2^{10} = 1024$$

$$\begin{aligned}
 \text{Total colours} &= 1024 \times 1024 \times 1024 \\
 &= (2^{10})^3 \\
 &= 2^{30} \\
 &= 1,073,741,824
 \end{aligned}$$

FINAL ANSWER: 1,073,741,824 possible colours per pixel (30 bits per pixel).

Source Grounding Used

- CSC514_Glossary_and_Answers.pdf – exact 2023/2024 Section A answer key and glossary definitions.
- CSC514_Q4_Q5_Solutions.pdf – full model solutions and calculations for Questions 4 and 5.
- CSC514_2022-2023_Full_Solutions.pdf – OpenGL, raster/vector, computer graphics applications, terminology and transformation explanations used where the 2023/2024 question revisits the same course concepts.
- CSC 514 LAST CLASS NOTE.pdf and CSC514 - Note 2.pdf – handwritten transformation and homogeneous-coordinate methods/diagrams.
- CSC 514 Color Models.pdf – RGB/CMY/CMYK explanations and colour-model formulas.

Note: where the 2023/2024 paper asks for an OpenGL code fragment rather than reproducing a code listing from the uploaded material, the code above is assembled directly from the OpenGL primitives/functions documented in the course materials (GLUT window creation, glBegin/glEnd, glColor and glVertex).